

# Ammar Ahmed

AI ENGINEER • COMPUTER SCIENCE & AI

6th of October, Giza, Egypt | +20 150 160 6307 | ammarahmedabdelmaboud17@gmail.com  
linkedin.com/in/ammara-bd-el-maboud | github.com/3vmmar

## SUMMARY

AI and Computer Science undergraduate with hands-on industry experience spanning AI engineering, large-scale model evaluation, and technical instruction. Skilled across deep learning, computer vision, NLP, and end-to-end ML pipelines — from data acquisition through deployment — with direct exposure to frontier AI systems and multimodal agent development.

## EDUCATION

**Zewail City of Science and Technology** — B.Sc. in Computer Science & Artificial Intelligence Sep 2023 – Present  
*Specialization in Data Science & Artificial Intelligence*

## EXPERIENCE

**AI Engineer — Anthropic** Mar 2026 – Present • Remote

- Contribute to model quality and AI safety initiatives at a leading frontier AI research lab
- Design evaluation pipelines to assess model capabilities, alignment, and behavioral robustness
- Support prompt engineering, behavioral analysis, and fine-tuning workflows for large language models

**Programming & Soft Skills Instructor — Udacity** Jul 2025 – Present • On Site

- Deliver project-based instruction covering Python, data science fundamentals, and professional development
- Design curriculum modules and mentor learners through end-to-end technical projects with structured feedback

**AI Model Evaluator — Alignerr** 2025 – Present • Remote

- Evaluate LLM outputs across reasoning, coding, and domain-specific tasks with structured, high-quality annotations
- Identify failure modes and edge cases in model behavior using domain expertise in AI and computer science

**Programming Instructor — iSchool Egypt** Aug 2024 – Present • Egypt

- Teach Python, AI, and machine learning concepts through project-based lessons that build real problem-solving skills

**AI Model Trainer — Outlier AI** Oct 2024 – Mar 2025 • Remote

- Fine-tuned and evaluated ML models using RLHF and instruction-following techniques
- Curated training datasets and delivered performance insights that directly improved model reliability

## PROJECTS

**IRIS — Multimodal AI Perception & Cognition Agent** 2026

*Python, YOLO12, DeepFace, Claude, Llama 3 / Ollama, FastAPI, WebSockets, ChromaDB, React 18*

- Built a real-time multimodal AI agent that perceives, remembers, and reasons over live camera and microphone input, integrating YOLO12 tracking, DeepFace identity recognition, and Faster-Whisper speech-to-text on an async FastAPI + WebSocket backend
- Engineered a hybrid cloud/local reasoning layer that switches between Claude and Llama 3 (via Ollama) for vision-language reasoning, backed by a ChromaDB vector-memory store and a semantic world model for spatial awareness
- Implemented a proactive rule-based agent for autonomous alerts and an immersive React 18 + Tailwind UI with real-time bounding-box overlays and neural text-to-speech

**ChestX-ray14 Pneumonia Detection with XAI** 2026

*PyTorch, EfficientNet, ResNet, DenseNet, ViT, GradCAM*

- Led a team deep-learning pipeline on the NIH ChestX-ray14 dataset (112K+ X-rays) to classify pneumonia
- Trained and evaluated 12 models including DenseNet121/201, Xception, ViT-B/16, EfficientNet-B3, and ResNet-101 with Test-Time Augmentation
- Applied 4 XAI methods per model (GradCAM, GradCAM++, ScoreCAM, EigenCAM) for clinically interpretable heatmaps; addressed class imbalance via focal loss and threshold tuning

**Arabic Text Generation — RNN / LSTM / Transformer** 2025

*PyTorch, LSTM, RNN, pyarabic, Wikipedia API*

- Designed and trained a comparative Arabic text-generation system across RNN, LSTM, and Transformer architectures at both character and word levels, achieving iterative perplexity improvements over 10+ experimental runs
- Engineered a full NLP pipeline covering Arabic Wikipedia scraping, Unicode normalization, BPE subword tokenization, and model evaluation

**Facial Analysis & Demographic Detection** 2024

*Python, MTCNN, DeepFace, OpenCV*

- Built a real-time multi-face detection and analysis system integrating MTCNN, DeepFace, and OpenCV
- Delivered age, gender, and emotion classification with bounding-box overlays and landmark localization

## TECHNICAL SKILLS

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| Languages            | Python, C/C++, SQL, JavaScript, C#, HTML/CSS                 |
| ML / Deep Learning   | PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, YOLO |
| Data & Visualization | Pandas, NumPy, Matplotlib, Seaborn, Plotly, OpenCV           |
| Backend & Web        | FastAPI, WebSockets, RESTful APIs, React, Node/JS            |
| Databases            | MySQL, PostgreSQL, MongoDB, Firebase, ChromaDB               |

|                    |                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloud & DevOps     | AWS, GCP, Azure, Docker, Git & GitHub                                                                                                                                          |
| AI / Research      | Deep Learning, Reinforcement Learning (RLHF), Large Language Models (LLMs), Multimodal / Agentic AI, Explainable AI (XAI), NLP, Computer Vision, Model Evaluation & Deployment |
| Languages (Spoken) | Arabic (Native) · English (Fluent)                                                                                                                                             |